CSE2525 · DATA MINING · 2026–27

# Distances & Similarity

*Choosing a distance is choosing which patterns an algorithm can see — worked through two case studies.*

A. Anand · study deck

PART I

# Why distance underlies data mining

*Before any formula: what does a distance actually let us do?*

THE ONE IDEA TO HOLD ONTO

# Choosing a distance is choosing what's similar

Every task today — classify, cluster, flag outliers, search — reduces to one question: what is close to what? Change the distance and the answer changes, though *no data point moved*.




"Similar" has **no default meaning** — it depends what you measure:

- Similar in price? — a running shoe lands next to a dress shoe.
- Similar in purpose? — it lands next to other running shoes.
- Similar in description text? — grouped by words, not by use.
- Similar in who buys it? — you get *complements*, not substitutes.

Same two candidates, same query — the nearest neighbour flips with the distance. The task decides which sense you want; the distance is how you encode it.

DISTANCE IS A SHARED BUILDING BLOCK

# One dataset — four questions




| Mining task | The question, expressed through distance |
|---|---|
| Classification | Which labelled examples are closest to this new object? |
| Clustering | Which objects are close enough to form groups? |
| Anomaly detection | Which objects are far from the usual observations? |
| Similarity search | Which existing objects are closest to this query? |

YOUR TURN · 2 MIN WITH YOUR NEIGHBOUR

# Design decision: the “more like this” button

SCENARIO

A music app wants a “more like this” button on every song. You decide what powers it.

— Which of today’s four questions is this — classification, clustering, anomaly, or similarity search?

— What should “similar” mean here — same genre, same tempo & mood, or the same listeners?

— Each answer is a different representation. Which one can you actually compute from the data you have?

There’s no correct answer — only a design decision you can defend. We’ll compare a few.

PART II

# Task → representation → distance

*The choice of distance is the model — so make the choice deliberately.*

“SIMILAR” IS AMBIGUOUS ON PURPOSE

# Close in what sense?

— Two shoes can be close in price but different in purpose.

— Two products can have similar descriptions but very different prices.

— Two visually similar products may be substitutes; two bought-together products may be complements.

CONSEQUENCE

Different features give different neighbours. Picking the feature and the distance *is* picking the model — this is the most important idea in the lecture, so we put it first.

BEFORE THE DISTANCE COMES THE ENCODING

# What is a *representation space*?

DEFINITION

A **representation** encodes each object as an element of a chosen space $\mathcal{R}$ — a vector, a set, a sequence, a node in a graph. Every later step sees **only** what $\mathcal{R}$ records, so two objects identical in $\mathcal{R}$ are **indistinguishable** to the distance. Choosing $\mathcal{R}$ is the **first** modelling decision.



Vector · $\mathbb{R}^n$

object → a point of numeric features

`Euclidean · cosine`



Set

object → a set of elements (words, shingles)

`Jaccard`

`1 0 1 1 0`

Sequence

object → an ordered string of symbols

`edit · Hamming`



Graph / tree

object → a node in a graph

`shortest path`

VECTOR SPACE · THE STRAIGHT-LINE DISTANCE

# Euclidean distance

$$d(x, y) = \sqrt{\left( \Sigma_i \, (x_i - y_i)^2 \right)}$$

WORKED EXAMPLE

x = (1, 2),  y = (4, 6).

d = $\sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(9 + 16)} = \sqrt{25} =$ **5**.

Cosine is its sibling for text — it compares *direction*, not magnitude. We'll meet both again: where they're used, and whether they're metrics.

SET · THE OVERLAP OF TWO SETS

# Jaccard distance

$$J(A, B) = |A \cap B| \,/\, |A \cup B| \qquad d_J = 1 - J(A, B)$$

WORKED EXAMPLE

$A = \{a, b, c, d\}$, $B = \{c, d, e\}$.

$A \cap B = \{c, d\} \to 2$, $A \cup B = \{a, b, c, d, e\} \to 5$. $J = 2/5 = 0.4$, $d_J =$ **`0.6`**.

Natural for documents seen as sets of words or shingles. We'll meet it again — in near-duplicate detection — and ask whether it's a metric.

SEQUENCE · HOW MANY EDITS APART

# Edit (Levenshtein) distance

*d(x, y) = fewest insertions, deletions & substitutions turning x into y*

WORKED EXAMPLE

x = `book`, y = `back`: b·b ✓, o → a, o → c, k·k ✓ → **2** substitutions.

Equal length, substitutions only → this coincides with the **Hamming** distance. Insertions & deletions let it compare *different* lengths: `book → books` = 1.

We'll meet Hamming again in Part III — where we prove it *is* a metric — and in the video case study.

GRAPH · THE FEWEST STEPS BETWEEN NODES

# Shortest-path (geodesic) distance

$$d(u, v) = \min \text{ over paths } u \to v \text{ of } \Sigma \text{ weight(edges)}$$

WORKED EXAMPLE

Edges: A–B = 1, B–C = 1, and a direct A–C = 5.

Paths A → C: direct = 5, via B = 1 + 1 = 2. So d(A, C) = **2** — the shortest path wins.

Unweighted graphs → just count edges. We'll work this out in full in Part III, and again in graph mining.

FOUR DECISIONS, IN ORDER

# Task → representation → distance → decision

| Choice | What it determines |
|---|---|
| Task | What “similar” should mean |
| Representation | What information is available for comparison |
| Distance | Which differences matter, and how much |
| Decision rule | What we do with the resulting distances (e.g. top-k) |

And you don’t always need a vector first — trees, strings and sets have distances directly.

YOUR TURN · WALK THE SEQUENCE

# Design decision: spot duplicate news articles

SCENARIO

A news aggregator scrapes the same story from dozens of sites, each with small edits. You must group the duplicates — fill in the four decisions.

| Decision | Your call for this task |
|---|---|
| Task | What should “similar” mean — near-identical wording? |
| Representation | The whole string? a bag of words? a set of word *k*-grams? |
| Distance | Edit distance? Jaccard on the sets? cosine on term vectors? |
| Decision rule | Group two articles when the distance is below which threshold? |

Notice the dependency: the representation you pick decides which distances are even available to you.

PART III

# Metrics as useful guarantees

*What properties make a distance predictable — and fast?*

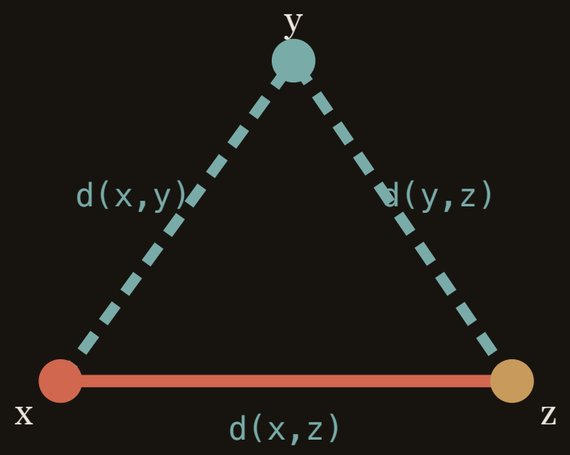
DEFINITION

# When is a distance a *metric*?

METRIC SPACE

A function $d : X \times X \to \mathbb{R}$ is a **metric** on a set $X$ if, for every $x, y, z \in X$, it satisfies the four axioms below. The pair **(X, d)** is then called a **metric space**.

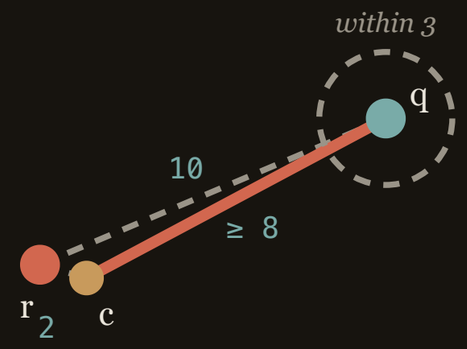
| Axiom | Formal statement |
|---|---|
| Non-negativity | $d(x, y) \geq 0$ |
| Identity of indiscernibles | $d(x, y) = 0 \iff x = y$ |
| Symmetry | $d(x, y) = d(y, x)$ |
| Triangle inequality | $d(x, z) \leq d(x, y) + d(y, z)$ |




Triangle inequality: the detour **x→y→z** is never shorter than going **direct**.

WHY THESE AXIOMS ARE WORTH DEMANDING

# The axioms are guarantees an algorithm can exploit




**c** is ≥ 8 from **q**, but we only want neighbours within 3 — so **c** is skipped, unmeasured.

— Identity and symmetry make “the nearest object” well-defined and independent of comparison order.

— The triangle inequality lets known distances bound unknown ones, so candidates can be ruled out *without* measuring them.

PRUNING WITH THE TRIANGLE INEQUALITY

*q* is at distance 10 from a reference *r*; a candidate *c* is at distance 2 from *r*. By the triangle inequality, **`d(q, c) ≥ |10 − 2| = 8`**. Seeking only neighbours within 3, we discard *c* **without ever computing d(q, c)**.

This powers metric indexes — ball trees, VP-trees — that avoid the full $O(n^2)$ comparisons.

EXERCISE · CHECK THE AXIOMS

# Is the Hamming distance a metric?




*2 positions differ, so the distance = 2*

CANDIDATE DISTANCE

For strings $x$, $y$ of equal length $n$, let $d_H(x, y)$ be the number of positions at which they differ.

— Verify each of the four axioms for $d_H$.

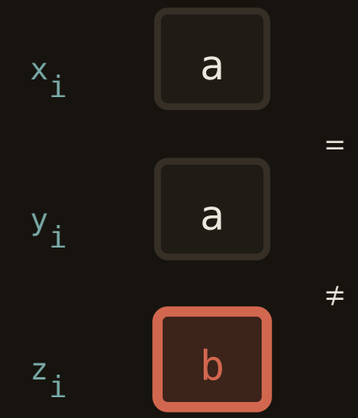
— Three are immediate; the triangle inequality takes an argument — reason position by position.

Decide before the next slide: is it a metric, and *why* does the triangle inequality hold?

Metric axioms to check: non-negativity $d(x,y) \geq 0$ · identity $d(x,y)=0 \Leftrightarrow x=y$ · symmetry $d(x,y)=d(y,x)$ · triangle $d(x,z) \leq d(x,y)+d(y,z)$

SOLUTION

# Yes — the Hamming distance is a metric




At one position: $x_i = y_i$ but $y_i \neq z_i$, so $x_i \neq z_i$ — at least one pair always differs.

PROOF — VERIFY EACH AXIOM

Non-negativity. $d_H$ is a count of positions, hence $d_H(x, y) \geq 0$.

Identity. $d_H(x, y) = 0 \iff$ no position differs $\iff x = y$.

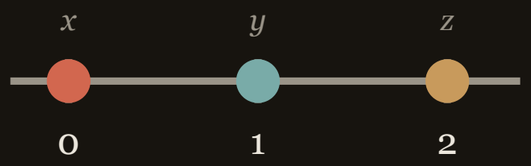
Symmetry. "position $i$ differs" is the same statement for (x, y) and (y, x), so $d_H(x, y) = d_H(y, x)$.

Triangle inequality. Fix a position $i$. If $x_i \neq z_i$, then $x_i$ and $z_i$ cannot both equal $y_i$, so $x_i \neq y_i$ *or* $y_i \neq z_i$. Hence at each position $[x_i \neq z_i] \leq [x_i \neq y_i] + [y_i \neq z_i]$. Summing over all $n$ positions gives $d_H(x, z) \leq d_H(x, y) + d_H(y, z)$.

∎

EXERCISE · FIND THE BROKEN AXIOM

# Is squared Euclidean distance a metric?




Compare the **direct** jump x→z against the **two-step** path x→y→z.

CANDIDATE DISTANCE

On $\mathbb{R}^n$, define $d_{sq}(x, y) = \|x - y\|^2 = \Sigma_i (x_i - y_i)^2$ — the **squared** Euclidean distance (the quantity *k*-means minimises).

— Three of the four axioms clearly hold. Which one can fail?

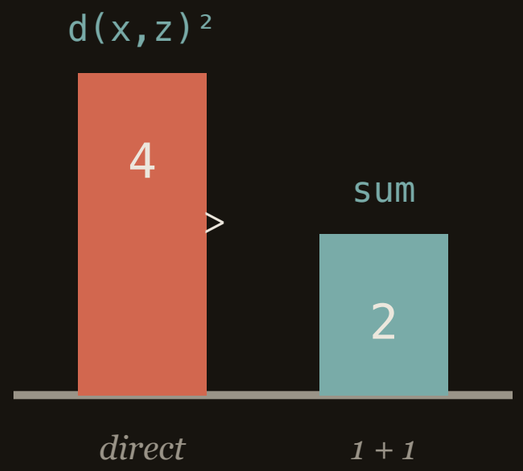
— Build a counterexample using the three points on the line.

Hint: compare the direct jump against the two-step path.

Metric axioms to check: non-negativity $d(x,y) \geq 0$ · identity $d(x,y)=0 \Leftrightarrow x=y$ · symmetry $d(x,y)=d(y,x)$ · triangle $d(x,z) \leq d(x,y)+d(y,z)$

SOLUTION

# No — the triangle inequality fails




Direct (**4**) > detour sum (**2**) — so **d(x,z) ≤ d(x,y)+d(y,z)** is broken.

COUNTEREXAMPLE ON THE REAL LINE

Take $x = 0$, $y = 1$, $z = 2$. Then

$d_{sq}(x, z) = (2 - 0)^2 = \mathbf{4}$, while $d_{sq}(x, y) + d_{sq}(y, z) = 1^2 + 1^2 = \mathbf{2}$.

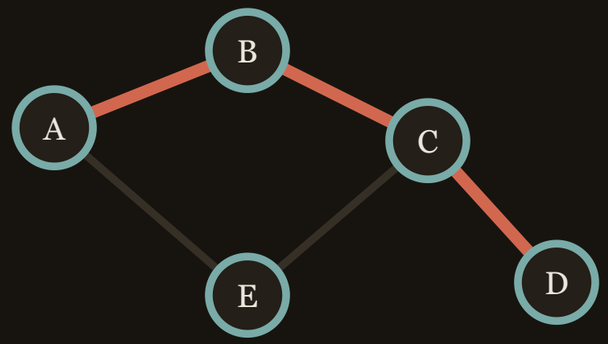
So $4 > 2$ — the triangle inequality is violated.

TAKEAWAY

Non-negativity, identity and symmetry all hold, yet $d_{sq}$ is not a metric. Squaring breaks the triangle inequality — which is why the Euclidean distance (its square root) is used wherever metric structure is required.

A METRIC ON OBJECTS THAT ARE NOT VECTORS

# Worked example: shortest-path distance on a graph




DEFINITION

In a connected graph, `d(u, v)` is the number of edges on a **shortest path** between $u$ and $v$ (the **geodesic** distance; for weighted graphs, the smallest total edge weight).

— BFS from A reads the distance straight off the layer number:

— `layer 0 = {A},  1 = {B, E},  2 = {C},  3 = {D}`.

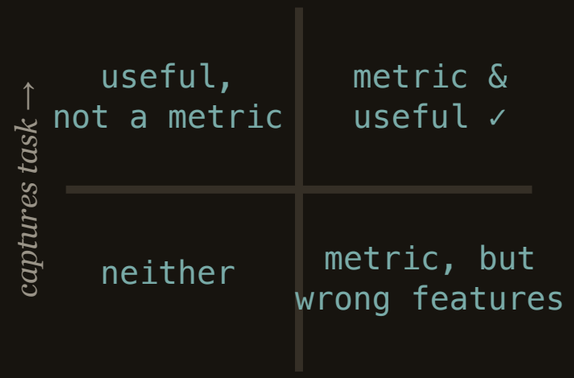
— So `d(A,D) = 3`, along A–B–C–D *or* A–E–C–D.

— Triangle check: `d(A,D) = 3 ≤ d(A,C) + d(C,D) = 2 + 1`. ✓

It is a metric: any path through a third node $w$ is itself a valid $u \to v$ path, so the shortest cannot be longer — the triangle inequality holds by construction.

KEEP THESE SEPARATE

# Valid metric ≠ useful distance




The two questions are **independent** — every box can be occupied.

ASK BOTH, INDEPENDENTLY

1 · Is it a mathematically valid metric?  2 · Does it capture the similarity our task needs?

Neither answer settles the other. A valid metric can be useless for a task; a genuinely useful dissimilarity need not satisfy every axiom. We'll see both in the case studies.

YOUR TURN · DO YOU NEED A METRIC?

# Design decision: 10 million face photos

SCENARIO

You must find near-duplicate faces in a 10-million-photo archive — far too many to compare every pair.

— To prune candidates without computing every distance, which axiom must your distance obey — and why that one?

— Your score gives d(A,B) ≠ d(B,A). Which axiom breaks, and does it actually matter here?

— Keep the two questions apart: is your score a valid metric, and does it capture “same person”?

The point of this part: metric properties aren’t pedantry — the triangle inequality is what makes the search feasible at all.

Metric axioms to check: non-negativity `d(x,y) ≥ 0` · identity `d(x,y)=0 ⇔ x=y` · symmetry `d(x,y)=d(y,x)` · triangle `d(x,z) ≤ d(x,y)+d(y,z)`

PART IV · CASE STUDY 1

# Products: which differences matter

*Designing a similarity, as a sequence of decisions.*

"I WANT TO SUGGEST SIMILAR ITEMS"

# First decide what relevance means




What makes a product a useful recommendation for another — shared category? co-purchase? "related items" / clicks vs. purchases define the target before any distance is chosen.

WHAT INFORMATION IS AVAILABLE

# Four representations of a product

### Category

- Where it sits in the catalogue tree

### Text

- Title & description terms

### Price

- A single numeric feature

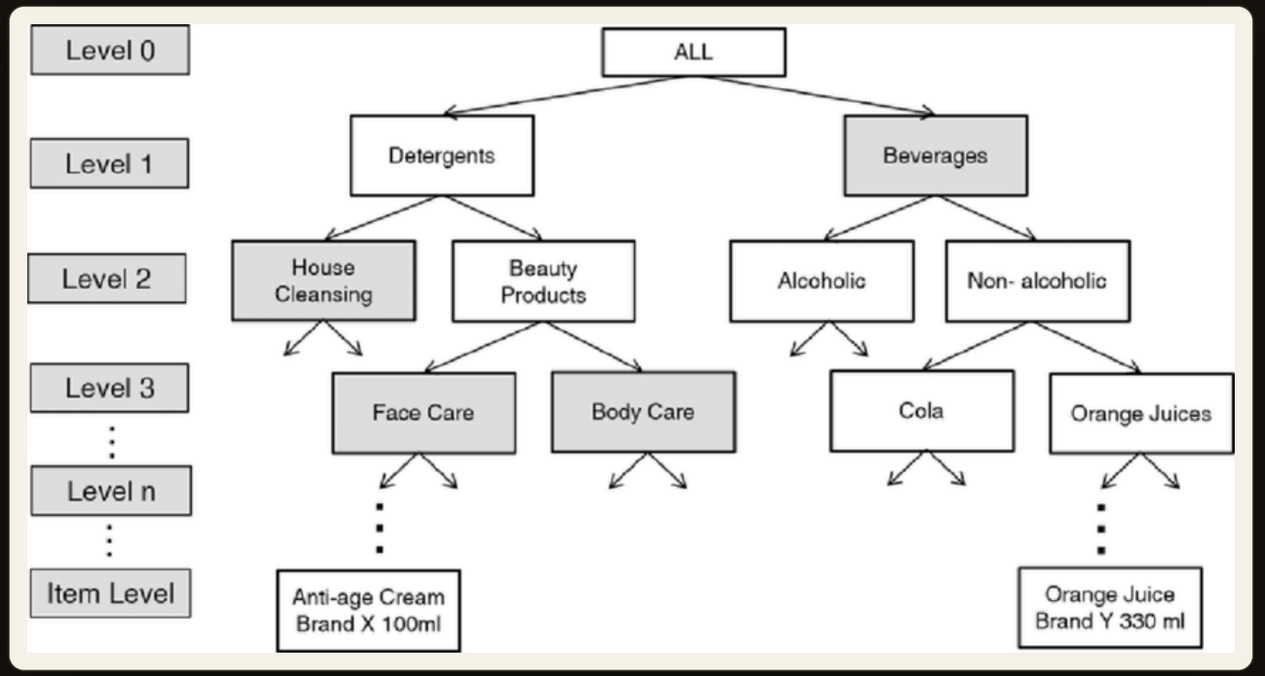
### Reviews

- Ratings & their distribution

Each representation needs its own distance — and each will disagree about who the neighbours are.

CATEGORY → A TREE

# Taxonomy distance = shortest path




Distance = path length up to a common ancestor and back down. A tree is a graph, so this is exactly the shortest-path distance from Part III — and it is a metric on category nodes. But distinct products in the same category have distance 0, so the representation, not the metric, is the limit.

TEXT → DIRECTION OF A TERM VECTOR

# Cosine: similar wording, any length

$$\cos(\theta) = (a \cdot b) / (\|a\| \, \|b\|) \quad d_{cos} = 1 - \cos(\theta)$$

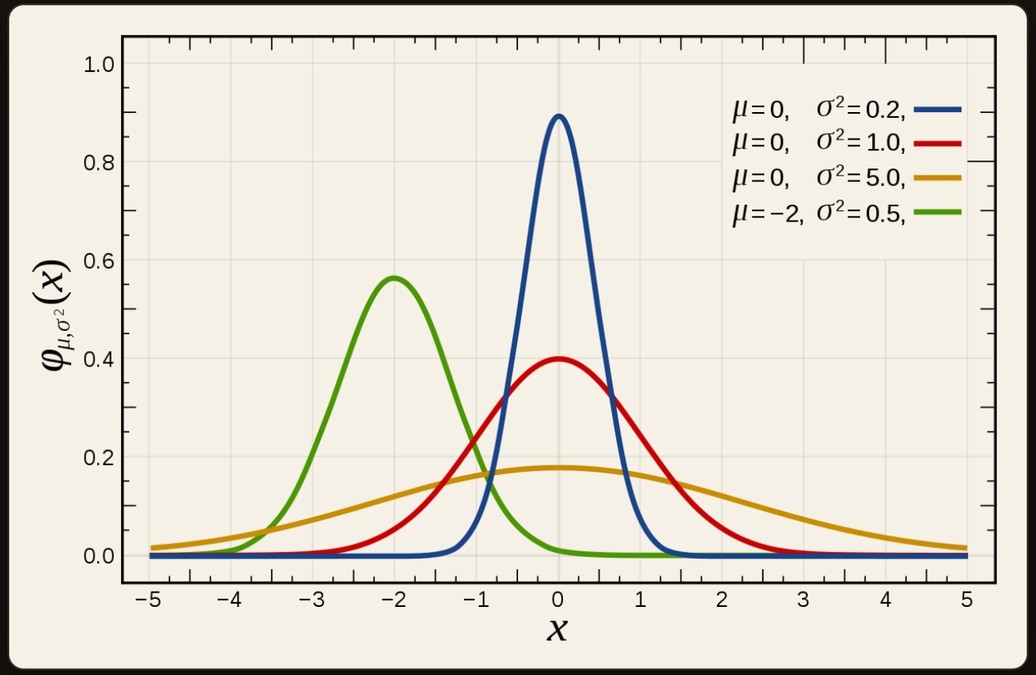
READ IT CAREFULLY

Only the direction of the term vectors matters, never their length — great for descriptions of different sizes.

Mind two traps: distance 0 means same direction, not identical (proportional vectors collapse to 0); and $1 - \cos\theta$ is not a true metric — it can violate the triangle inequality. Useful, but not a metric.

PRICE → EUCLIDEAN, BUT BEWARE UNITS

# Standardize — then don’t over-trust it




Raw price dominates small-scale features, so standardize: $z = (x - \mu)/\sigma$. But equal scale ≠ equal relevance, and it does not remove outlier sensitivity — scaling is not a priority decision.

REVIEWS → SPARSE & COLD-START

# What about products with no history?




Most products have few reviews; brand-new ones have none (cold start). The distance must degrade gracefully when a representation is missing or thin.

QUESTION

Your query is a **running shoe**. Ranked by **price**, the single closest product is a formal **dress shoe**. Is that a good recommendation — and what decides the answer?

*Same three candidates, three different distances.*

Think — then → for the answer.

SEE THE CHOICE CHANGE THE ANSWER

# One query, three candidates, three rankings

QUERY

a **running shoe** · €120 · *Footwear › Running* · “lightweight breathable running shoe”.

| Candidate | By price | By category | By text |
|---|---|---|---|
| A · trail running shoe (€130) | near (Δ10) | nearest — same “Running” | near — “running shoe” |
| B · running socks (€15) | far (Δ105) | far — Apparel › Socks | near — “breathable lightweight running” |
| C · formal dress shoe (€125) | nearest (Δ5) | mid — shares “Footwear” | far — only “shoe” |

Price crowns a dress shoe; text pulls in socks; only the right representation ranks the other running shoe first. The distance is the model.

FROM OBJECTIVE TO ALGORITHM

# Combine distances — weights are priorities

$$d(x,y) = w_c \cdot d_{category} + w_t \cdot d_{text} + w_p \cdot d_{price}$$

TWO CAUTIONS, LINKED

The terms must be on comparable scales before you add them — exactly the standardization point again.

The weights w encode priorities; they are the model's hyperparameters. Standardization does not choose them for you — you decide what should matter. Then rank the top-k neighbours and evaluate against real clicks/purchases.

PART V · CASE STUDY 2

# Videos: which differences to ignore

*Invariance, thresholds, and a system that scales.*

YOUTUBE: NEAR-DUPLICATES OF AN UPLOAD

# Equivalent for our task, different in the bytes



Figure 1. Examples of queries and near-duplicate videos from CC_WEB_VIDEO dataset.

What should be ignored: re-encoding, modest colour changes, logos/overlays, small crops. The job is the opposite of case 1 — choosing differences to discard.

BYTE EQUALITY IS THE WRONG TOOL

# Exact ≠ near

THE FAILURE

Byte equality and **cryptographic hashes** (MD5, SHA) are designed so one changed pixel flips the whole digest. They detect **identical files**, never **perceptually similar** ones.

We need a representation where small perceptual changes → small distance.

CONSTRUCT THE RIGHT REPRESENTATION

# Perceptual hashing (pHash)

THE GENERAL IDEA

Reduce each frame to a short fingerprint that survives re-encoding and minor edits — typically from low-frequency structure, so cosmetic changes don’t move it.

Keep the idea separate from any one scheme. Code length, keyed behaviour and non-invertibility are properties of *particular* hashes, not defining features of perceptual hashing in general.

COMPARE → HAMMING DISTANCE

# Count the differing bits

$a = 10110010 \quad b = 10111010 \rightarrow d_{Hamming} = 1$

BACK TO PART III

Hamming distance is a proper metric on fingerprints — we verified all four axioms in Part III.

Yet distinct videos can share a fingerprint. A metric *in the representation space* does not guarantee the originals are distinguishable — the same lesson as taxonomy-distance on products.

MAKE A DECISION, ACCEPT ERRORS

# Pick a threshold — trade off the two mistakes

DECISION RULE

call it a near-duplicate when Hamming distance ≤ **τ**.

- Threshold too tight → miss real duplicates (false negatives).
- Threshold too loose → flag unrelated videos (false positives).
- τ is chosen from data — exactly the kind of knob the task, not the metric, decides.

MAKE IT SCALE

# Fast filter → slow check

*cheap candidate generation → expensive exact verification*

A RECURRING PATTERN — FAST FILTER, SLOW CHECK

Generate a small candidate set cheaply (fingerprint buckets / LSH), then verify the few survivors with the expensive comparison. Never materialise all $n^2$ distances.

The same shape powers web search, retrieval and question answering — a recurring pattern this year.

SYNTHESIS

# Change the distance, change the patterns

*Why this one choice ripples through every task.*

BACK TO THE FOUR TASKS

# Changing the distance changes the answer

- Nearest neighbours change → classifications and recommendations change.
- Neighbourhoods change → clusters and density estimates change.
- What counts as “far” changes → detected anomalies change.

| Case | Main lesson |
| --- | --- |
| Products | Choosing which differences matter |
| Videos | Choosing which differences to ignore |

TAKE AWAY · AND A TRANSFER EXERCISE

# Choosing a distance is choosing what you can see

THE MESSAGE

Task → representation → distance → decision. Metrics buy predictability and speed; but a valid metric and a task-useful distance are two separate questions.

Choosing a distance is choosing which patterns an algorithm can see.

Transfer: for detecting duplicate news articles — what representation would you choose, which changes should be ignored, and how would you evaluate the result?